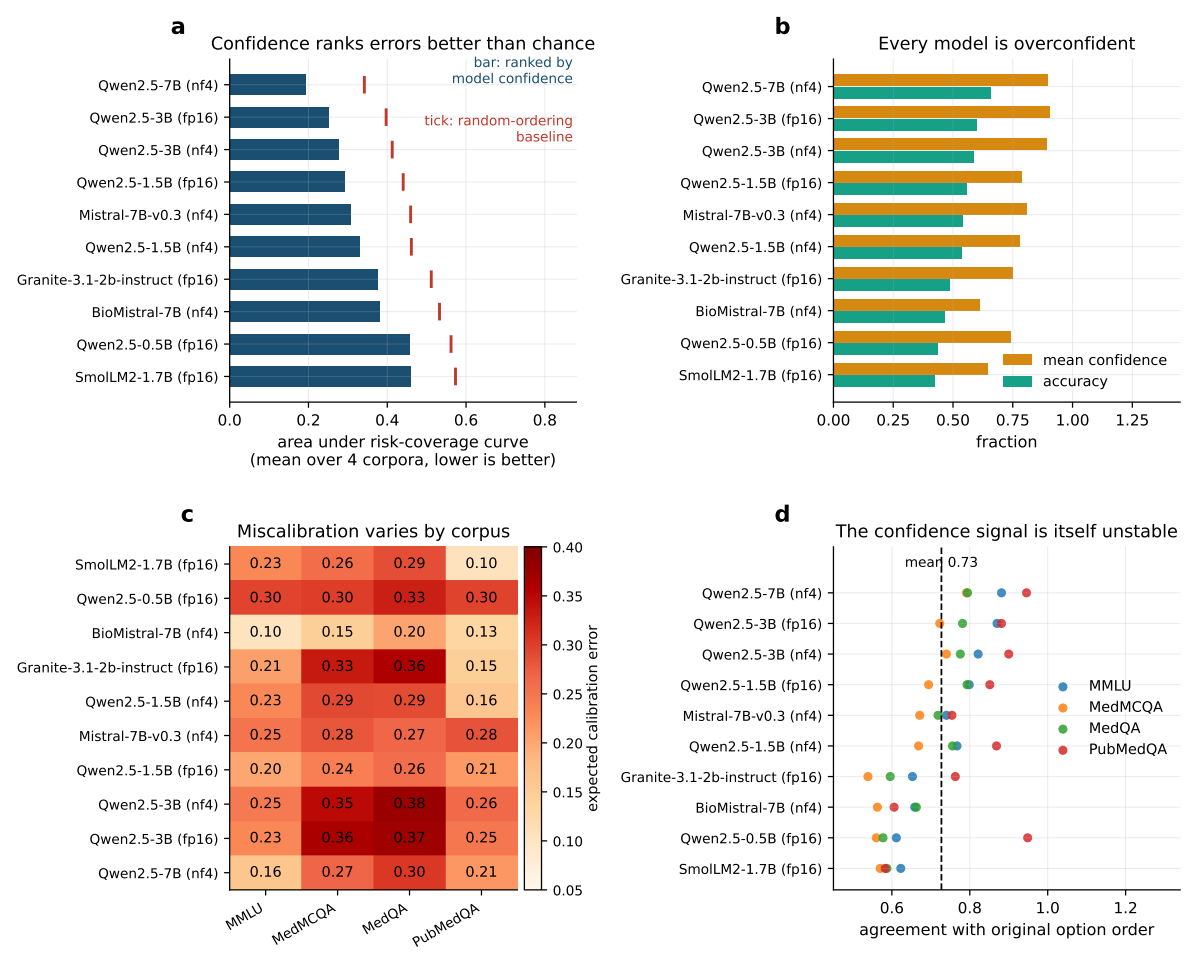

a
Confidence ranks errors better than chance
bar: ranked by model confidence
tick: random-ordering baseline
Qwen2.5-7B (nf4)
Qwen2.5-3B (fp16)
Qwen2.5-3B (nf4)
Qwen2.5-1.5B (fp16)
Mistral-7B-v0.3 (nf4)
Qwen2.5-1.5B (nf4)
Granite-3.1-2b-instruct (fp16)
BioMistral-7B (nf4)
Qwen2.5-0.5B (fp16)
SmolLM2-1.7B (fp16)
0.0 0.2 0.4 0.6 0.8
area under risk-coverage curve
(mean over 4 corpora, lower is better)
b
Every model is overconfident
mean confidence
accuracy
0.00 0.25 0.50 0.75 1.00 1.25
fraction
c
Miscalibration varies by corpus
SmolLM2-1.7B (fp16) 0.23 0.26 0.29 0.10
Qwen2.5-0.5B (fp16) 0.30 0.30 0.33 0.30
BioMistral-7B (nf4) 0.10 0.15 0.20 0.13
Granite-3.1-2b-instruct (fp16) 0.21 0.33 0.36 0.15
Qwen2.5-1.5B (nf4) 0.23 0.29 0.29 0.16
Mistral-7B-v0.3 (nf4) 0.25 0.28 0.27 0.28
Qwen2.5-1.5B (fp16) 0.20 0.24 0.26 0.21
Qwen2.5-3B (nf4) 0.25 0.35 0.38 0.26
Qwen2.5-3B (fp16) 0.23 0.36 0.37 0.25
Qwen2.5-7B (nf4) 0.16 0.27 0.30 0.21
MMLU
MedMCQA
MedQA
PubMedQA
expected calibration error
0.40 0.35 0.30 0.25 0.20 0.15 0.10 0.05
d
The confidence signal is itself unstable
mean 0.73
MMLU
MedMCQA
MedQA
PubMedQA
0.6 0.8 1.0 1.2
agreement with original option order